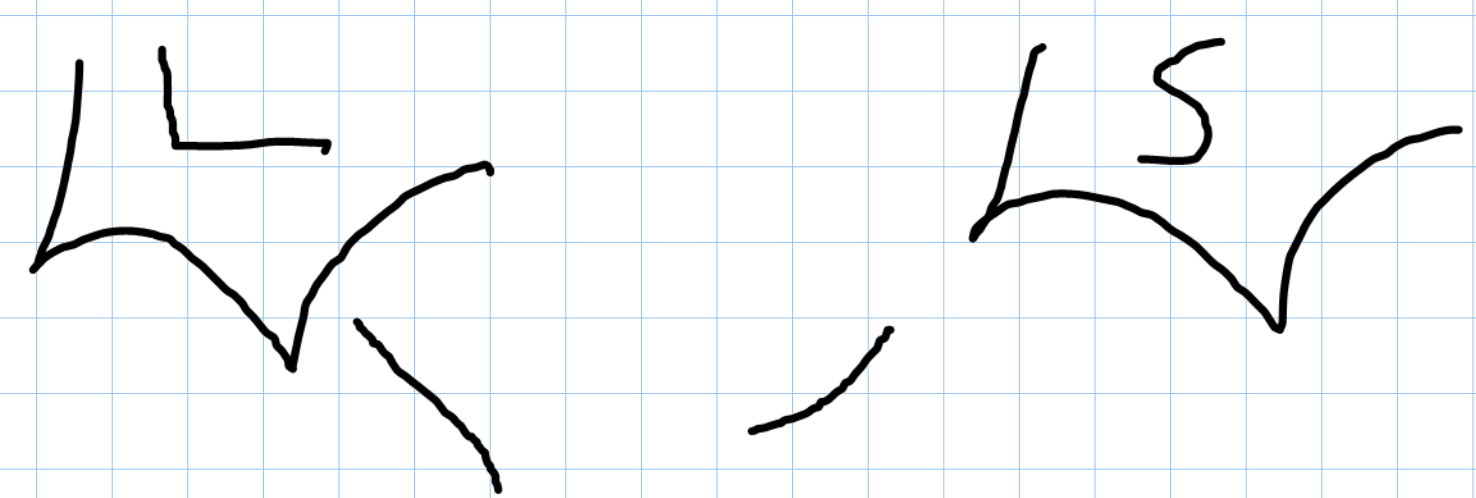




$$L = L_1 + L_2 \quad S = S_1 + S_2 \quad [L, S] = 0$$

$$J = (L + S) \Rightarrow J^2 = L^2 + S^2 + 2LS$$

$$LS = L_x S_x + L_y S_y + L_z S_z$$

$$L_+ = L_x + iL_y \quad S_+ = S_x + iS_y$$

$$L_- = L_x - iL_y \quad S_- = S_x - iS_y$$

$$L_+ S_- + L_- S_+$$

$$L_x S_x + i(L_y S_x - L_x S_y) + L_y S_y$$

$$+ L_x S_x + i(L_x S_y - L_y S_x) + L_y S_y$$

$$2L_x S_x + 2L_y S_y = L_+ S_- + L_- S_+$$

$$\Rightarrow L_x S_x + L_y S_y = \frac{1}{2} (L_+ S_- + L_- S_+)$$

$$J^2 = L^2 + S^2 + 2LS$$

$$L^2 + S^2 + 2L_z + L_+ S_- + L_- S_+$$

$$\Psi = \alpha \chi_+ \psi_{lm} \chi_-$$

$$\psi_{lm} \chi_+ + \psi_{l, m+1} \chi_-$$

$$\alpha \psi_{lm} \chi_+ + \beta \psi_{l, m+1} \chi_-$$

$$L^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle \quad L_z |l, m\rangle = m\hbar |l, m\rangle$$

$$S^2 |s, m_s\rangle = \frac{3}{4} \hbar^2 |s, m_s\rangle \quad S_z |s, m_s\rangle = m_s \hbar |s, m_s\rangle \quad \left. \begin{array}{l} \\ \end{array} \right\} L_z S_z$$

$$S_+ \chi_- = \hbar \chi_+ \quad S_+ \chi_+ = 0$$

$$S_- \chi_+ = \hbar \chi_- \quad S_- \chi_- = 0$$

$$L^2 \Rightarrow \alpha \hbar^2 \{ l(l+1) \psi_{lm} \chi_+ \} + \beta \hbar^2 \{ l(l+1) \psi_{l, m+1} \chi_- \}$$

$$S^2 \Rightarrow \alpha \hbar^2 \{ \frac{3}{4} \psi_{lm} \chi_+ \} + \beta \hbar^2 \{ \frac{3}{4} \psi_{l, m+1} \chi_- \}$$

$$2L_z S_z \Rightarrow \alpha \hbar^2 \{ \frac{m}{2} \psi_{lm} \chi_+ \} + \beta \hbar^2 \{ -\frac{m+1}{2} \psi_{l, m+1} \chi_- \}$$

$$L_+ S_- \Rightarrow \alpha \hbar^2 \{ \sqrt{l(l+1)-m(m+1)} \psi_{l, m+1} \chi_- \} + \beta$$

$$L_- S_+ \Rightarrow \quad \quad \quad + \beta \hbar^2 \{ \sqrt{l(l+1)-m(m-1)} \psi_{lm} \chi_+ \}$$

$$\hbar^2 \psi_{lm} \chi_+ \left\{ \alpha \left(l(l+1) + \frac{3}{4} + m \right) + \beta \sqrt{l(l+1)-m(m-1)} \right\} = J(J+1) \hbar^2 \psi_{lm} \chi_+$$

$$\hbar^2 \psi_{l, m+1} \chi_- \left\{ \alpha \sqrt{l(l+1)-m(m+1)} + \beta \left(l(l+1) + \frac{3}{4} - m - 1 \right) \right\} = J(J+1) \hbar^2 \psi_{l, m+1} \chi_-$$

$$\frac{1}{2} \psi_{lm} \chi_+ \left\{ \alpha \left(l(l+1) + \frac{3}{4} + m \right) + B \sqrt{l(l+1) - m(m-1)} \right\} = j(j+1) \frac{1}{2} \psi_{lm} \chi_+$$

$$\frac{1}{2} \psi_{lm+1} \chi_- \left\{ \alpha \sqrt{l(l+1) - m(m+1)} + B \left(l(l+1) + \frac{3}{4} - m - 1 \right) \right\} = j(j+1) \frac{1}{2} \psi_{lm+1} \chi_-$$

$$B \sqrt{l(l+1) - m(m-1)} = \left(j(j+1) - l(l+1) - \frac{3}{4} - m \right)$$

$$d = j(j+1) - l(l+1) - \frac{3}{4} \quad \begin{matrix} \nearrow l \\ \searrow l-1 \end{matrix}$$

$$\alpha \sqrt{l(l+1) - m(m+1)} = \left(j(j+1) - l(l+1) - \frac{3}{4} + m + 1 \right)$$

$$j = l + \frac{1}{2} \Rightarrow d = l$$

$$j = l - \frac{1}{2} \Rightarrow d = l - 1$$

$$d = \left(l - \frac{1}{2} \right) \left(l + \frac{1}{2} \right) - l^2 - l - \frac{3}{4}$$

$$l^2 - \frac{1}{4} - l^2 - l - \frac{3}{4} = -l - 1$$

$$d = l;$$

$$\alpha \sqrt{l(l+1) - m(m+1)} = l + m + 1 \Rightarrow \alpha = \frac{l + m + 1}{\sqrt{l^2 + l - m^2 - m}}$$

$$B = \frac{l - m}{\sqrt{l^2 + l - m^2 + m}}$$

$$\frac{l - m}{\sqrt{l^2 + l - m^2 + m}}$$

$$\frac{l - m}{\sqrt{(l+m)(l-m+1)}}$$

$$\frac{l + m + 1}{\sqrt{(l-m)(l+m+1)}}$$

$$(l-m)(l+m+1) = l(l+1) - m(m+1)$$

$$\text{CVP } \alpha = \frac{\sqrt{l+m+1}}{\sqrt{2l+1}}$$

$$B = \frac{\sqrt{l-m}}{\sqrt{2l+1}}$$